Pre-Characterization Process Risk Assessment

A-Mab Drug Substance — A-Mab Drug Substance — RA-001

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

Table of contents

Field	Value
Document ID	RA-001
Document class	Pre-Characterization Process Risk Assessment
Title	Pre-Characterization Process Risk Assessment: A-Mab Drug Substance
Product	A-Mab
Version	1.0
Effective date	2026-07-24

Field	Value
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; BLA, Biologics License Application; CEX, cation exchange chromatography; CofA, certificate of analysis; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; IgG1, immunoglobulin G subclass 1; IPC, in-process control; KPP, key process parameter; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; pCO2, dissolved carbon dioxide; RPN, risk priority number; RRF, risk ranking and filtering; SEC, size exclusion chromatography; UF/DF, ultrafiltration and diafiltration; WC-CPP, well controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

The Pre-Characterization Process Risk Assessment decides which A-Mab process parameters are characterized, and by what kind of study. Because the assessment is performed before any characterization study is executed, the output is a scope and not a result: each of the 37 process parameters of the drug-substance train leaves this document with an assigned study type and with the reason for that assignment recorded against it.

Three limits on that output hold throughout. No parameter is classified in this assessment, because the critical, well controlled critical, key and general process parameter designations are outputs of the characterization studies themselves; the reports PCR-003 to PCR-010 assign them under the decision logic of the case study (CMC Biotech Working Group 2009). No residual risk is calculated for those parameters either. Since the control strategy that would reduce the risk is not yet defined, what is recorded here is the risk as it stands under the controls that exist today. Finally, no acceptable range is set here. Ranges are proposed for study (Appendix A), while a proven acceptable range is an output of the study for the step (International Council for Harmonisation 2009).

PTP-001, the Process Transfer Plan, is the parent of this assessment and fixes the transfer scope within which it works, while the consumers are the eight Process Characterization Plans (PCP-003 to PCP-010), each of which carries out the scope defined here for one unit operation. Above those eight plans, the campaign is covered by PCMP-001. The whole package is listed in Section 7.

The assessment covers Steps 3 to 10 of the drug-substance process, from the production bioreactor (Step 3) through to ultrafiltration and diafiltration (Step 10). The train is set out with the principal role of each step in Table 1.1. Seed expansion and the drug product process lie outside the scope, because the seed steps accumulate no product and have been shown to pose a low risk to product quality (CMC Biotech Working Group 2009). Equipment, raw materials and analytical methods are assessed in their own documents, which are referred to here only where the existing control of a parameter rests on one of them.

Table 1.1: The A-Mab drug-substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA

Step	Unit operation	Principal role
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

2 Risk assessment methodology

Risk ranking and filtering (RRF) is the method (CMC Biotech Working Group 2009). Each process parameter is scored on the potential impact of a deviation on a quality attribute (the main effect) and on the potential of that parameter to interact with other parameters of the same unit operation; the study type follows from the two scores together. Throughout, criticality is treated as a continuum: neither an attribute nor a parameter is placed in a binary critical or non-critical box at this stage (International Council for Harmonisation 2023b).

Severity is taken from the attribute a failure mode can reach, because a parameter inherits the severity of the most critical attribute within reach. Therefore a parameter that could move high mannose is scored at the top of the scale whatever the size of the effect later turns out to be, while a parameter that reaches only the acidic charge variants is scored near the bottom. The severity of a parameter never depends on how well a facility controls it, which keeps the criticality of an attribute separate from the capability of a plant (CMC Biotech Working Group 2009). The scale is given in Table 2.1, whose five bands run from a deviation with no impact on product quality to one that would have a lot rejected.

Table 2.1: Severity scale. The severity of a parameter is the severity of the most critical attribute its failure mode can reach.

Score	Band	Definition
10	Very High	Definite impact to product quality; lot(s) rejected
8	High	Probable impact; significant supplemental testing / re-processing
6	Moderate	Potential impact; supplemental t=0 testing / minor re-processing
4	Slight	Unlikely impact; no supplemental testing / no re-processing
2	Low/None	No impact to product quality

Occurrence and detection are scored as they stand before characterization. That is what makes the resulting number an initial one. Occurrence is 7 for a parameter that can reach a quality attribute, because neither the size of the effect nor the acceptable range is yet known; for a parameter that acts on process performance alone the score falls to 4. Detection is scored at 8 where a deviation would reach product quality without being visible in process. Where the attribute at risk is viral clearance, detection takes the lowest band of the scale at 10, because clearance is demonstrated in small-scale spiking studies and is never measured on the batch. For the performance-only parameters, whose deviations show up in the next operation, detection is scored at 6 (both scales are given in Table 2.2).

Table 2.2: Occurrence and detection scales, scored as they stand before characterization.

Score	Occurrence band	Occurrence definition	Detection band	Detection definition
10	Very High	Failure ~ once per 100 units	None	Not detected by in-process or CofA testing
8	High	Failure ~ once per 1,000 units	Low	Not detected in-process; caught by CofA testing
6	Moderate	Failure ~ once per 2,000 units	Moderate	Detected downstream before CofA, not during the unit op
4	Low	Failure ~ once per 5,000 units	High	Detected in the next downstream operation
2	Minimal	Failure ~ once per 10,000 units or lower	Very High	Immediately detected by in-process monitoring / inspection

The three scales together give the initial risk priority number, which is their product on an ordinal scale. The number therefore ranks parameters against one another and carries no further meaning; a ratio between two of the numbers says nothing about how much more risk one of them represents.

The study type is decided by the two axes of the ranking. A parameter whose deviation could take a quality attribute outside its acceptance criterion, and which prior knowledge places in interaction with another parameter of the same step, is studied in a multivariate design. A quality-linked parameter that acts through a single mechanism, or whose response is measured in a separate study, is assessed one factor at a time under a written justification, while a parameter with no credible route to any quality attribute is assessed one factor at a time to support process performance. The method also permits a fourth outcome, in which a parameter of negligible risk is not studied at all (CMC Biotech Working Group 2009); Section 5 records what this assessment did with that outcome.

The same ranking is summarised more coarsely as a priority band: High for a quality-linked parameter whose severity reaches 8, Medium for a quality-linked parameter below that severity, and Low for a parameter that acts on process performance alone. The three bands are filled by 16, 5 and 16 parameters respectively.

The same three scales are applied again after characterization, when each report weighs the residual risk and designates the critical process parameters of its own step (CMC Biotech Working Group 2009). That later step of the lifecycle lies outside this document. None of the thresholds used in that step is applied to any number in this document.

3 Quality attributes at risk

The attribute register is the severity axis of this assessment. The 10 attributes the drug substance must meet are listed in Table 3.1, with the acceptance criterion, the criticality assigned in the product risk assessment, the Tool #1 score behind that criticality, the severity each attribute confers on a parameter that can reach it and the step at which the attribute is set.

Table 3.1: Quality attributes, criticality and the severity each confers.

Quality attribute	Category	Acceptance	Criticality	Tool #1	Severity	Set by
Afucosylation	glycosylation	2–13 % of glycans	H	60	8	Production Bioreactor
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48	8	Production Bioreactor
High Mannose	glycosylation	3–10 % of glycans	VH	80	10	Production Bioreactor
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60	8	Production Bioreactor
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4	4	Production Bioreactor
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36	6	Production Bioreactor
Residual DNA	process impurity	0–0.001 ng/dose	VL	6	2	Production Bioreactor
Leached Protein A	process impurity	0–5 ppm	L-M	16	4	Protein A Chromatography
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20	10	Low-pH Viral Inactivation
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20	10	Anion Exchange Chromatography

3 of the attributes carry very high criticality: high mannose, cumulative XMuLV clearance and cumulative MVM clearance. High mannose reaches the highest Tool #1

score in the register at 80, because the attribute carries both a high impact on activity and real uncertainty about the size of that impact (CMC Biotech Working Group 2009). By contrast, the two viral clearance attributes (XMuLV and MVM) score 20, well below the glycan attributes, because the mechanism of clearance is understood and very little uncertainty attaches to it. However, the severity all three confer is the same, at 10, since a batch that misses any one of the three cannot be released.

Below that band the attribute register spreads widely in severity, from high criticality down to very low. Afucosylation, galactosylation and aggregates (HMW) sit at high criticality and confer a severity of 8. Host cell protein (HCP) follows at moderate to high criticality, with three chromatography steps (Protein A, cation exchange and anion exchange) sharing its clearance. The lowest of the register are the acidic charge variants (deamidation) and residual DNA, both of very low criticality, of which residual DNA confers the lowest severity of all at 2.

7 of the 10 attributes are formed at the production bioreactor, which is why that step carries the longest parameter list in the assessment. Leached Protein A enters at the capture step, while the two viral clearance attributes are credited to the step that carries the larger part of each claim: the low-pH hold (Step 6) for XMuLV and anion exchange (Step 8) for MVM. All three viral steps contribute to both claims, with the cumulative figures consolidated in PCMR-001 (International Council for Harmonisation 2023a).

Both of those claims rest on an acceptance criterion the process must reach from below. Cumulative XMuLV and MVM clearance are stated as minimum log reductions. A deviation that lowers clearance is therefore the failure of interest at Steps 6, 8 and 9, while a deviation that raises it is of no concern.

Severity travels from an attribute to a parameter through a postulated failure mode. No attribute in that register is re-scored here. The register is taken as it stands from the product risk assessment, where impact and uncertainty were the only two axes used and neither manufacturing capability nor detectability entered the score (CMC Biotech Working Group 2009). What this assessment adds is the reach of each parameter, one unit operation at a time.

4 Parameter risk ranking by unit operation

The assessment identified a credible route to a quality attribute for 21 of the 37 parameters. The remaining 16 act on process performance only (titre, yield, filter capacity and buffer exchange). The highest initial risk number in the assessment, 700, is reached by 8 parameters, every one of which belongs to one of Steps 6, 8 and 9 (the three steps credited with viral clearance). Detection drives that result. Nothing in it says that those three steps are poorly controlled.

The parameters of each unit operation are given below, with the failure mode postulated for each, the attributes that failure mode could reach and the resulting score, all read against the scales of Section 2. The two steps whose parameters reach no attribute at all are correspondingly short.

4.1 Production bioreactor (Step 3)

Because the production bioreactor forms more quality attributes than any other step in the process, the parameter list of the step is correspondingly long: 9 parameters enter the assessment, of which 5 can reach a quality attribute.

Table 4.1: Pre-characterization risk ranking, Production Bioreactor.

Parameter	Potential failure mode	Attribute(s) at risk	Potential effect	Sev.	Initial RPN
Culture pH	pH excursion outside the characterized range (6.6-7.1)	Afucosylation, Galactosylation (total %Gal), High Mannose, Acidic charge variants (deamidation), Aggregates (HMW)	Shifts glycosylation (ADCC via afucosylation; CDC via galactosylation) and charge/aggregate profile	10	560
Culture temperature	Temperature excursion outside 34-36 C	Afucosylation, Galactosylation (total %Gal), Acidic charge variants (deamidation)	Alters afucosylation, galactosylation and acidic charge variants; interacts with pH	8	448
Dissolved CO2 (pCO2)	Dissolved CO2 outside 40-160 mmHg	Galactosylation (total %Gal), Acidic charge variants (deamidation)	Strong effect on galactosylation and acidic variants	8	448

Parameter	Potential failure mode	Attribute(s) at risk	Potential effect	Sev.	Initial RPN
Osmolality	Osmolality outside 360-440 mOsm	Galactosylation (total %Gal), Acidic charge variants (deamidation)	Affects galactosylation and acidic variants	8	448
Culture duration	Culture extended beyond 15-19 days / harvested on low viability	Afucosylation, Galactosylation (total %Gal), Aggregates (HMW), Host Cell Protein (HCP), Residual DNA	Lowers afucosylation & galactosylation; raises HCP/DNA and aggregate as viability declines	8	448
Dissolved oxygen	DO outside 30-70%	process performance	Impacts peak VCD, titer and process consistency (no significant CQA impact)	4	96
Initial viable cell conc.	Inoculation density outside 0.5-1.5e6/mL	process performance	Drives peak/integral VCD and titer; no product-quality impact	4	96
Basal medium concentration	Basal medium concentration outside 0.8-1.6X	process performance	Statistically significant but shallow effect on afucosylation; drives titer	4	96
Nutrient feed-1 volume	Nutrient feed-1 volume outside 9-15% WV	process performance	Slight effect on galactosylation; critical for titer	4	96

Culture pH is ranked highest of the group at 560 (Table 4.1). A pH excursion reaches 5 of the attributes formed in the culture, high mannose among them, which places the parameter at the top of the severity scale. Culture duration reaches the same number of attributes at a lower severity, because the two it adds are host cell protein and residual DNA, both of which clear downstream at lower criticality than any glycan attribute.

Culture temperature, dissolved carbon dioxide and osmolality follow at 448, each reaching galactosylation and the acidic charge variants, with temperature reaching afucosylation as well. Since prior knowledge places an interaction between culture pH and temperature (CMC Biotech Working Group 2009), the 5 quality-linked parameters of the step are studied together in one design and not one at a time.

The four remaining parameters of the step act on cell growth, titre and process consistency without reaching any quality attribute. Dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume are all ranked at 96, the floor of the assessment. Two of those four parameters need a word of care: prior knowledge records a small but statistically significant effect of basal medium concentration on afucosylation, together with a slight effect of nutrient feed-1 volume on galactosylation (CMC Biotech Working Group 2009). Both effects are shallow enough that the case study treated neither as a quality risk, which is why both parameters are ranked here at the performance band. However, the univariate studies assigned to them in Section 5 will bound both effects across the ranges proposed for study. That is the check that would overturn the ranking.

4.2 Harvest and clarification (Step 4)

No parameter of the harvest step can change a product quality attribute. The step removes cells and cell debris from the culture, while the antibody itself passes through unmodified (CMC Biotech Working Group 2009). All 3 parameters of the step are therefore ranked at the floor of the assessment, at 96.

Table 4.2: Pre-characterization risk ranking, Harvest and Clarification.

Parameter	Potential failure mode	Attribute(s) at risk	Potential effect	Sev.	Initial RPN
Centrifugation (rcf)	Centrifugation rcf outside 6000-12000 g	process performance	Affects clarification/yield (no CQA impact)	4	96
Depth filter load	Depth filter load outside 80-160 L/m2	process performance	Affects filter capacity/turbidity (no CQA impact)	4	96
Post-clarification turbidity	Post-clarification turbidity high	process performance	Feed quality to Protein A (no CQA impact)	4	96

Centrifugation speed and depth filter load are both sized by the clarification duty, with post-clarification turbidity as the in-process control (IPC) that reports the outcome of both. The clarified harvest is the feed to the capture step, where feed quality is assessed as an input to that step.

4.3 Protein A chromatography (Step 5)

Of the 6 parameters at the capture step (Step 5), two reach a quality attribute. The attribute is host cell protein in both cases.

Table 4.3: Pre-characterization risk ranking, Protein A Chromatography.

Parameter	Potential failure mode	Attribute(s) at risk	Potential effect	Sev.	Initial RPN
Protein load	Protein load above characterized range (>50 g/L resin)	Host Cell Protein (HCP)	Increases pool HCP (interaction with elution pH)	6	336
Elution buffer pH	Elution pH below characterized range (<3.2)	Host Cell Protein (HCP)	Increases pool HCP; also sets up viral inactivation feed	6	336
Load flow rate	Load flow rate high with high load	process performance	Reduces product yield (no CQA impact)	4	96
End of pool collect	End-of-pool collection outside 2.0-3.2 CV	process performance	Affects product yield (no CQA impact)	4	96
Operating temperature	Parameter outside characterized range	process performance	No significant product-quality impact expected	4	96
Bed height	Parameter outside characterized range	process performance	No significant product-quality impact expected	4	96

Protein load and elution buffer pH are ranked at 336. Loading above the range proposed for study leaves more host cell protein in the pool, at a level that depends on elution pH (CMC Biotech Working Group 2009). That interaction is why both parameters are carried into a multivariate design. Neither of them carries the whole burden for host cell protein: because cation exchange (Step 7) and anion exchange clear the impurity further downstream, the acceptable output of the capture step is set with those two steps in view (CMC Biotech Working Group 2009).

Elution buffer pH does a second job at this step, since the pH of the Protein A eluate sets up the low-pH hold that follows. Therefore the parameter is also an input to the viral inactivation assessed at Step 6, which is why the range proposed for study here was chosen to keep that hold inside its own acceptable window.

Load flow rate and end of pool collect act on yield alone and are ranked at the performance floor, at 96. Both are nevertheless carried into the multivariate design of the step, for the reason given in Section 5. Operating temperature and bed height carry no specific failure mode into the assessment, because equipment design and the column packing criteria hold both parameters, neither of which is expected to reach a quality attribute.

4.4 Low-pH viral inactivation (Step 6)

The low-pH hold is the first of the three steps that carry a viral clearance claim. Of its 4 parameters, 3 reach a quality attribute, all three of them at 700, the highest number in the assessment.

Table 4.4: Pre-characterization risk ranking, Low-pH Viral Inactivation.

Parameter	Potential failure mode	Attribute(s) at risk	Potential effect	Sev.	Initial RPN
Inactivation pH	pH > 4.0 (incomplete inactivation) or pH < 3.2 (aggregation/precipitation)	Viral clearance — XMuLV (enveloped), Aggregates (HMW)	Loss of XMuLV inactivation assurance or increased aggregate	10	700
Hold time	Hold time < 60 min (incomplete) or > 180 min (aggregation)	Aggregates (HMW), Viral clearance — XMuLV (enveloped)	Incomplete inactivation or increased aggregate	10	700
Temperature	Temperature < 15 C	Viral clearance — XMuLV (enveloped)	Lower inactivation rate at short time / higher pH	10	700
A-Mab concentration	Concentration > 35 g/L	process performance	No significant effect on inactivation, aggregation or charge variants	4	96

The governing parameter is inactivation pH, whose failure mode runs in two directions. Above the range proposed for study the hold has not been shown to inactivate XMuLV within the hold time, while below that range the antibody begins to aggregate and to precipitate (CMC Biotech Working Group 2009). Hold time fails in the same two directions, since a short hold leaves inactivation incomplete and a long hold raises aggregate at the low pH of the step. Temperature acts in one direction only, because a cold hold inactivates more slowly at a given time and pH.

A-Mab concentration is the one parameter of the step that reaches nothing, because prior knowledge across the platform shows no effect of concentration on inactivation kinetics, on aggregation or on the charge variants (CMC Biotech Working Group 2009). The parameter ranks at 96 and is assessed one factor at a time.

Detection is what lifts this step above the bioreactor. Because viral clearance is never measured on a manufacturing batch, detection is scored at the bottom of the scale for every parameter that can reach a clearance attribute. That detection score records the absence of a batch level check and no weakness in the control of pH, hold time or temperature, all three of which are held by verified titration, a timed hold with alarms and a monitored ambient temperature.

4.5 Cation exchange chromatography (Step 7)

Cation exchange is the principal aggregate polishing step of the train. Because no step downstream of it reduces aggregate, the aggregate risk of the whole purification train concentrates here (CMC Biotech Working Group 2009). Of the 5 parameters of the step, 4 reach a quality attribute.

Table 4.5: Pre-characterization risk ranking, Cation Exchange Chromatography.

Parameter	Potential failure mode	Attribute(s) at risk	Potential effect	Sev.	Initial RPN
Protein load	Protein load above characterized range (>40 g/L resin)	Aggregates (HMW), Host Cell Protein (HCP)	Reduced aggregate and HCP clearance (interactions with pH and wash conductivity)	8	448
Load/Wash conductivity	Load/wash conductivity below 3 mS/cm	Host Cell Protein (HCP)	Reduced HCP clearance	6	336
Elution buffer pH	Elution pH outside 5.8-6.2	Aggregates (HMW)	Alters aggregate distribution and yield	8	448
Elution stop collect	Extended elution stop-collect (>1.5 OD)	Aggregates (HMW)	Minor increase in pool aggregate	8	448
Elution flow rate	Elution flow rate outside 100-300 cm/hr	process performance	Affects peak shape and yield (minor aggregate)	4	96

Protein load is ranked highest at 448, because the failure mode reaches both aggregate and host cell protein. Elution buffer pH and elution stop collect are ranked at the same number through aggregate alone, while load and wash conductivity reaches host cell protein only and is ranked at 336. Since prior knowledge places interactions between protein load, elution pH and wash conductivity, those four are studied together in one design (CMC Biotech Working Group 2009). Elution flow rate affects peak shape and yield, ranks at the performance floor and is assessed one factor at a time.

The input aggregate to this step is bounded by the low-pH hold above it, because a high aggregate feed would push the step towards its acceptance criterion at any load. Input aggregate is therefore treated as a material attribute of the step and not as one of its parameters, with PCP-007 setting the feed states at which the study is run.

4.6 Anion exchange chromatography (Step 8)

Every parameter of the anion exchange step reaches a quality attribute, because the step polishes host cell protein in flow-through mode and carries the larger part of the cumulative MVM claim (CMC Biotech Working Group 2009). Two very high criticality attributes and one moderate to high attribute therefore sit within reach of the same 5 parameters.

Table 4.6: Pre-characterization risk ranking, Anion Exchange Chromatography.

Parameter	Potential failure mode	Attribute(s) at risk	Potential effect	Sev.	Initial RPN
Load pH	Load pH below characterized range (<7.2)	Host Cell Protein (HCP), Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	Reduced HCP removal and reduced XMuLV/MVM clearance	10	700
Equil/Wash-1 conductivity	Equil/Wash-1 conductivity above 3.6 mS/cm	Host Cell Protein (HCP)	Reduced HCP removal	6	336
Load conductivity	Load conductivity above characterized range	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	Reduced viral clearance (interacts with pH)	10	700
Protein load	Protein load above characterized range (>300 g/L resin)	Host Cell Protein (HCP)	Reduced HCP removal capacity (viral clearance robust)	6	336
Operating flow rate	Operating flow rate above 450 cm/hr	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	Reduced residence time / viral clearance margin	10	700

Load pH is ranked at 700 and reaches the widest set of attributes at the step, which is host cell protein together with both viral clearance attributes. Load conductivity and operating flow rate are also ranked at 700 through the two viral attributes, while equilibration and wash-1 conductivity and protein load reach host cell protein alone and are ranked at 336.

Prior knowledge places an interaction between load pH and load conductivity on viral clearance, with clearance falling as pH falls and conductivity rises (CMC Biotech

Working Group 2009). Operating flow rate is the one parameter in the whole assessment that ranks at 700 and is still assigned a univariate study; Section 5 gives the justification.

4.7 Small-virus retentive filtration (Step 9)

Small-virus retentive filtration removes virus by size and is orthogonal in mechanism to the two viral steps above it (International Council for Harmonisation 2023a). Both of the 2 parameters of the step are ranked at 700, the top of the assessment.

Table 4.7: Pre-characterization risk ranking, Small-Virus Retentive Filtration.

Parameter	Potential failure mode	Attribute(s) at risk	Potential effect	Sev.	Initial RPN
Filtration volume (load)	Volumetric load above 105 L/m2	Viral clearance — MVM (parvovirus)	Reduced MVM log-reduction (parvovirus breakthrough risk)	10	700
Filtration pressure	Pressure outside filter operating limits	Viral clearance — MVM (parvovirus), Viral clearance — XMuLV (enveloped)	Potential impact on virus removal at high load	10	700

The parameter of record is volumetric load. Prior knowledge shows the MVM log reduction of this filter type falling slowly as the load volume rises (CMC Biotech Working Group 2009). Breakthrough becomes the concern at the top of the proposed range, which is why the failure mode is written against the volumetric load including the chase (the total volume passed through the filter). Filtration pressure reaches both viral attributes. That parameter is held within the operating limits of the filter manufacturer and is checked by an integrity test before and after use.

4.8 Ultrafiltration and diafiltration (Step 10)

No parameter of the ultrafiltration and diafiltration step forms or clears a quality attribute. Because the step concentrates the drug substance and exchanges the buffer as a formulation duty, all 3 parameters are ranked at the floor of the assessment.

Table 4.8: Pre-characterization risk ranking, Ultrafiltration / Diafiltration.

Parameter	Potential failure mode	Attribute(s) at risk	Potential effect	Sev.	Initial RPN
Number of diavolumes	Diavolumes < 7	process performance	Incomplete buffer exchange (formulation; not part of DS characterization)	4	96
Transmembrane pressure	TMP outside 10-15 psi	process performance	Affects flux/processing (formulation)	4	96
Final DS concentration	Final concentration outside 65-85 g/L	process performance	DS concentration spec (formulation)	4	96

Diavolume count, transmembrane pressure and final concentration are controlled for process performance and for the concentration of the drug substance. Final concentration is verified against the drug substance release test, while the other two are verified in process.

5 Characterization study assignment

The output of the assessment is the assignment in Table 5.2, which gives every parameter of the train with the study type it will receive: 22 parameters are assigned to a multivariate design, 14 to univariate assessment and 1 to a justified univariate assessment. No parameter is excluded from study. The same split by unit operation, with the plan and the report that cover each step, is given in Table 5.1.

Table 5.1: Characterization scope by unit operation, with the covering documents.

Step	Unit operation	Parameters	Multivariate	Univariate	Documents
3	Production Bioreactor	9	5	4	PCP-003 / PCR-003
4	Harvest / Clarification	3	0	3	PCP-004 / PCR-004
5	Protein A Chromatography	6	4	2	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	1	PCP-006 / PCR-006
7	Cation Exchange Chromatography	5	4	1	PCP-007 / PCR-007
8	Anion Exchange Chromatography	5	4	1	PCP-008 / PCR-008
9	Small Virus Retentive Filtration	2	2	0	PCP-009 / PCR-009
10	Ultrafiltration / Diafiltration (formulation)	3	0	3	PCP-010 / PCR-010

6 of the 8 steps in the assessment are given a multivariate design, at Steps 3, 5, 6, 7, 8 and 9. The two steps without one are harvest and UF/DF, neither of which reaches a quality attribute.

A parameter is placed in the multivariate group when a deviation could take a quality attribute outside its acceptance criterion and when prior knowledge places an interaction with another parameter of the same unit operation. Both conditions matter. A parameter with a large main effect and no interaction can be bounded one factor at a time. A parameter that interacts cannot be bounded that way, because the acceptable range of one parameter then depends on the level of the other (International Council for Harmonisation 2009).

Two parameters in the multivariate group reach no quality attribute at all. Load flow rate and end of pool collect at the capture step act on yield. Because the design of that step already varies protein load and elution buffer pH across the runs it needs for host cell protein, adding the two performance parameters to the same design costs no further runs and resolves their interaction with protein load in the same analysis.

The univariate group is made up of the 14 parameters with no credible route to a quality attribute. Each of them is still studied. Because a range that supports the batch record has to come from somewhere, the cheapest source of one is a study at scale-down that moves a single factor at a time (a univariate ranging study). In addition, a null result from such a study is informative in its own right, since it keeps the parameter in the knowledge space as evidence of robustness that a later change to the process can draw on (U.S. Food and Drug Administration 2011).

One parameter sits between the two groups. Operating flow rate at anion exchange reaches both viral clearance attributes and is ranked at 700, yet the assignment is nevertheless a justified univariate study. Viral clearance is measured in dedicated small-scale spiking studies and not in the process design of experiments (DoE), because the spiking work needs containment and a different analytical method (International Council for Harmonisation 2023a). Since flow rate acts on clearance through residence time alone, which is a single mechanism with no interaction to resolve, the parameter is bounded by a flow rate series inside the spiking study. The justification is recorded in PCP-008.

The method permits a fourth outcome, in which a parameter of negligible risk is not studied at all (CMC Biotech Working Group 2009). No A-Mab parameter is placed there. Because the train is small and the scale-down systems are established, a ranging study on a low-risk parameter costs little against the value of a documented range.

Table 5.2: Characterization study assignment for every parameter.

Unit operation	Parameter	Sev.	Initial RPN	Priority	Study type
Production Bioreactor	Culture pH	10	560	High	multivariate DoE
Production Bioreactor	Culture temperature	8	448	High	multivariate DoE
Production Bioreactor	Dissolved CO2 (pCO2)	8	448	High	multivariate DoE
Production Bioreactor	Osmolality	8	448	High	multivariate DoE
Production Bioreactor	Culture duration	8	448	High	multivariate DoE
Production Bioreactor	Dissolved oxygen	4	96	Low	univariate
Production Bioreactor	Initial viable cell conc.	4	96	Low	univariate
Production Bioreactor	Basal medium concentration	4	96	Low	univariate

Unit operation	Parameter	Sev.	Initial RPN	Prior-ity	Study type
Production Bioreactor	Nutrient feed-1 volume	4	96	Low	univariate
Harvest and Clarification	Centrifugation (rcf)	4	96	Low	univariate
Harvest and Clarification	Depth filter load	4	96	Low	univariate
Harvest and Clarification	Post-clarification turbidity	4	96	Low	univariate
Protein A Chromatography	Protein load	6	336	Medium	multivariate DoE
Protein A Chromatography	Elution buffer pH	6	336	Medium	multivariate DoE
Protein A Chromatography	Load flow rate	4	96	Low	multivariate DoE
Protein A Chromatography	End of pool collect	4	96	Low	multivariate DoE
Protein A Chromatography	Operating temperature	4	96	Low	univariate
Protein A Chromatography	Bed height	4	96	Low	univariate
Low-pH Viral Inactivation	Inactivation pH	10	700	High	multivariate DoE
Low-pH Viral Inactivation	Hold time	10	700	High	multivariate DoE
Low-pH Viral Inactivation	Temperature	10	700	High	multivariate DoE
Low-pH Viral Inactivation	A-Mab concentration	4	96	Low	univariate
Cation Exchange Chromatography	Protein load	8	448	High	multivariate DoE
Cation Exchange Chromatography	Load/Wash conductivity	6	336	Medium	multivariate DoE
Cation Exchange Chromatography	Elution buffer pH	8	448	High	multivariate DoE
Cation Exchange Chromatography	Elution stop collect	8	448	High	multivariate DoE
Cation Exchange Chromatography	Elution flow rate	4	96	Low	univariate
Anion Exchange Chromatography	Load pH	10	700	High	multivariate DoE
Anion Exchange Chromatography	Equil/Wash-1 conductivity	6	336	Medium	multivariate DoE
Anion Exchange Chromatography	Load conductivity	10	700	High	multivariate DoE

Unit operation	Parameter	Sev.	Initial RPN	Prior-ity	Study type
Anion Exchange Chromatography	Protein load	6	336	Medium	multivariate DoE
Anion Exchange Chromatography	Operating flow rate	10	700	High	univariate (justified)
Small-Virus Retentive Filtration	Filtration volume (load)	10	700	High	multivariate DoE
Small-Virus Retentive Filtration	Filtration pressure	10	700	High	multivariate DoE
Ultrafiltration / Diafiltration	Number of diavolumes	4	96	Low	univariate
Ultrafiltration / Diafiltration	Transmembrane pressure	4	96	Low	univariate
Ultrafiltration / Diafiltration	Final DS concentration	4	96	Low	univariate

Three bounds apply to this assignment. The assignment rests on postulated failure modes and on platform knowledge alone; no effect recorded in it has been measured at A-Mab scale-down. Each assignment also covers only the range proposed for study (Appendix A), and a wider range would need a fresh assessment. Finally the assignment is prospective, since each Process Characterization Plan may amend it before execution and the report that follows may place a parameter differently.

Two kinds of amendment are expected. Both are handled by the plan for the step. A screening study may find no resolvable effect for a parameter carried into a multivariate design, in which case the parameter stays in the knowledge space as evidence of robustness and the report for the step classifies it on that basis. However, a univariate study may instead find an effect the assessment did not anticipate, in which case the plan for that step is re-issued before execution, which is an amendment to the plan and not a deviation from it.

6 Assumptions and limitations

Four assumptions carry the assessment. Each assumption can fail in a way the studies would expose.

The first assumption is platform transferability. A-Mab is developed on the humanized IgG1 platform that produced X-Mab, Y-Mab and Z-Mab, from which lineage the mechanisms recorded in this document are taken (CMC Biotech Working Group 2009). Where a mechanism does not carry across, the failure mode written here is the wrong one for A-Mab. The error would be revealed by the study assigned to that parameter.

The second assumption is the completeness of the parameter list. The 37 parameters were taken from the process description of the drug-substance train. A parameter absent from that description is not assessed here, while a parameter added later through change control needs this assessment repeated for it.

The third assumption is the scale-down model. Because every study assigned in Section 5 will be run at scale-down, the assignment presumes a model that represents the commercial process for the attribute in question. Qualification of each model is covered by the plan for that step and is not covered here (U.S. Food and Drug Administration 2011).

The fourth assumption is the scoring itself. Occurrence is scored from the uncertainty of the effect and not from failure frequency data, because commercial-scale data for A-Mab do not exist yet. Detection is scored on what a manufacturing batch would reveal, which is why the viral parameters take the lowest band despite a well controlled hold and a validated filter.

Two limitations follow from the method itself. The risk priority number is ordinal, which means that a ratio between two of the numbers carries no meaning and that only the rank order is used in Section 5. The second limitation is the absence of a residual figure. No residual risk is calculated anywhere in this document. The control strategy that would reduce the risk is defined by the characterization studies and consolidated in PCMR-001. The residual assessment belongs there.

7 Outputs and downstream use

The assignment of Section 5 is issued to the Process Characterization Plans. Each of those plans takes the parameters of one unit operation, with the study type assigned here, and sets out the design, the sampling plan and the acceptance criteria for that study. Above those eight plans, the campaign as a whole is covered by PCMP-001, which states the methods and the acceptance framework common to all of them.

The reports return the result. Those reports, PCR-003 to PCR-010, record the studies and classify the parameters of their own step, while PCMR-001 consolidates the classifications and the control strategy for the drug substance. This assessment is superseded for a given parameter once the report that classifies that parameter is approved. The whole package, with the subject of each document, is listed in Table 7.1.

Table 7.1: The A-Mab process characterization document package.

Document ID	Document class	Subject
PTP-001	Process Transfer Plan	A-Mab Drug Substance
PCMP-001	Process Characterization Master Plan	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)

Document ID	Document class	Subject
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance

8 Approvals

The signatories in Table 1 approve this assessment. Approval covers the method of Section 2, the ranking of Section 4 and the study assignment of Section 5.

Appendix A — Set-points, proposed study ranges and existing controls

Table 8.1 gives the set-point, the range proposed for study and the existing control for every parameter in the assessment. The ranges are proposals, since a proven acceptable range is an output of the characterization study and appears in the report for the step.

Table 8.1: Set-point, range proposed for study and existing control, by parameter.

Step	Parameter	Set-point	Range proposed for study	Existing control
3	Culture pH	6.85 pH	6.6–7.1 pH	Closed-loop pH control with validated/redundant probes; continuous in-process monitoring; multivariate design space 6.6-7.1
3	Culture temperature	35 °C	34–36 °C	Jacket temperature control; validated probes; in-process monitoring; design space 34-36 C
3	Dissolved CO2 (pCO2)	100 mmHg	40–160 mmHg	Sparge/overlay gas control; pCO2 monitoring; design space 40-160 mmHg
3	Osmolality	400 mOsm	360–440 mOsm	Feed/base osmolality control; at-line monitoring; design space 360-440 mOsm
3	Culture duration	17 days	15–19 days	Harvest criteria on viability, titer and quality; duration bounded 15-19 days
3	Dissolved oxygen	50 % sat	30–70 % sat	Cascade DO control; in-process monitoring
3	Initial viable cell conc.	1 e6 cells/mL	0.5–1.5 e6 cells/mL	Seed-train control; split-ratio spec
3	Basal medium concentration	1.2 X	0.8–1.6 X	Media prep controls; concentration verified
3	Nutrient feed-1 volume	12 % WV	9–15 % WV	Gravimetric feed control; feed schedule
4	Centrifugation (rcf)	9000 g	6000–12000 g	Centrifuge speed controlled; turbidity IPC

Step	Parameter	Set-point	Range proposed for study	Existing control
4	Depth filter load	120 L/m2	80–160 L/m2	Sizing by area; turbidity IPC
4	Post-clarification turbidity	5 NTU	1–20 NTU	Turbidity IPC; Protein A robust to feedstock
5	Protein load	30 g/L resin	10–50 g/L resin	Load controlled by A280/mass balance; design space 10-50 g/L; CEX/AEX downstream HCP clearance
5	Elution buffer pH	3.55 pH	3.2–3.9 pH	Buffer pH verified pre-use; in-line pH monitoring; design space 3.2-3.9
5	Load flow rate	200 cm/hr	100–300 cm/hr	Flow controlled by pump/skid; yield IPC
5	End of pool collect	2.6 CV	2–3.2 CV	Collection by CV/A280; yield IPC
5	Operating temperature	22 °C	15–30 °C	Standard equipment control
5	Bed height	20 cm	10–30 cm	Standard equipment control
6	Inactivation pH	3.5 pH	3.2–4 pH	pH adjust to 3.5 +/- 0.1 with verified titration; in-line pH; acceptable range 3.2-4.0
6	Hold time	90 min	60–180 min	Timed hold 60-120 min with alarms; CEX downstream aggregate polish
6	Temperature	21 °C	15–25 °C	Ambient hold 15-25 C; temperature monitored
6	A-Mab concentration	20 g/L	5–35 g/L	Pool concentration <= 35 g/L (typically <= 20 g/L)
7	Protein load	25 g/L resin	10–40 g/L resin	Load by mass balance; design space 10-40 g/L; input aggregate bounded by VI
7	Load/Wash conductivity	5 mS/cm	3–7 mS/cm	Buffer conductivity verified; in-line conductivity; design space 3-7 mS/cm
7	Elution buffer pH	6 pH	5.8–6.2 pH	Buffer pH verified; in-line pH; design space 5.8-6.2
7	Elution stop collect	1 OD desc.	0.5–1.5 OD desc.	Stop-collect by descending A280; design space 0.5-1.5 OD
7	Elution flow rate	200 cm/hr	100–300 cm/hr	Flow controlled by skid
8	Load pH	7.5 pH	7.2–7.8 pH	Load pH adjust to 7.5; in-line pH; quality design space 7.2-7.8; viral clearance validated >= 7.0

Step	Parameter	Set-point	Range proposed for study	Existing control
8	Equil/Wash-1 conductivity	2.6 mS/cm	1.6–3.6 mS/cm	Buffer conductivity verified; in-line conductivity; design space 1.6-3.6 mS/cm
8	Load conductivity	5.5 mS/cm	3–8 mS/cm	Conditioned load conductivity; viral clearance validated <= 15 mS/cm
8	Protein load	200 g/L resin	50–300 g/L resin	Load by mass balance; design space 50-300 g/L; viral clearance validated to 300 g/L
8	Operating flow rate	300 cm/hr	150–450 cm/hr	Flow controlled by skid; viral clearance validated <= 450 cm/hr
9	Filtration volume (load)	90 L/m2	50–140 L/m2	Volumetric load limited to <= 105 L/m2 incl. chase; filter integrity test
9	Filtration pressure	13 psi	8–30 psi	Pressure within manufacturer limits; monitored; pre/post integrity test
10	Number of diavolumes	8 DV	7–10 DV	Diavolume count controlled; buffer exchange IPC
10	Transmembrane pressure	12 psi	10–15 psi	TMP controlled by skid
10	Final DS concentration	75 g/L	65–85 g/L	Concentration by UV/mass; release test

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.

International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.